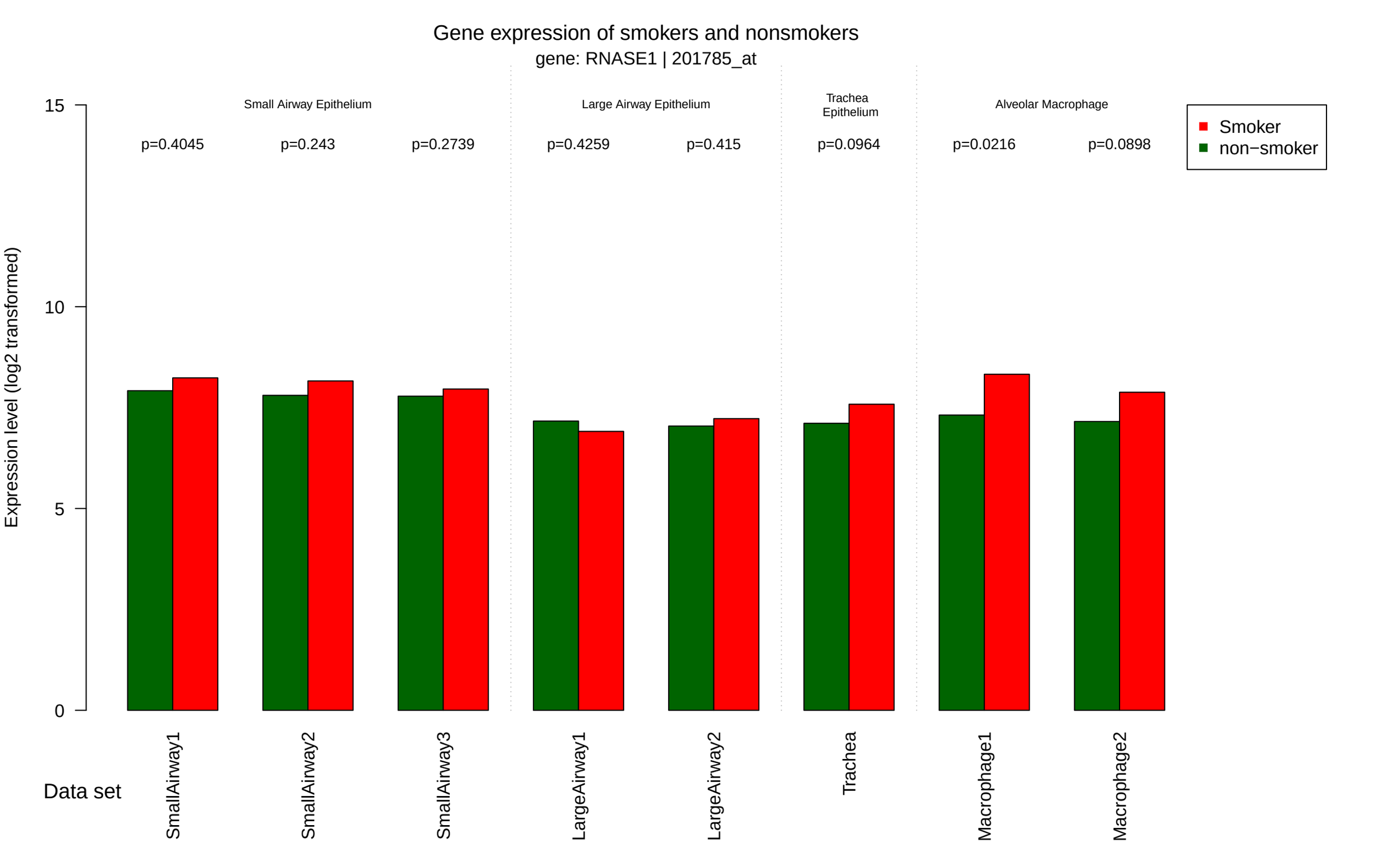
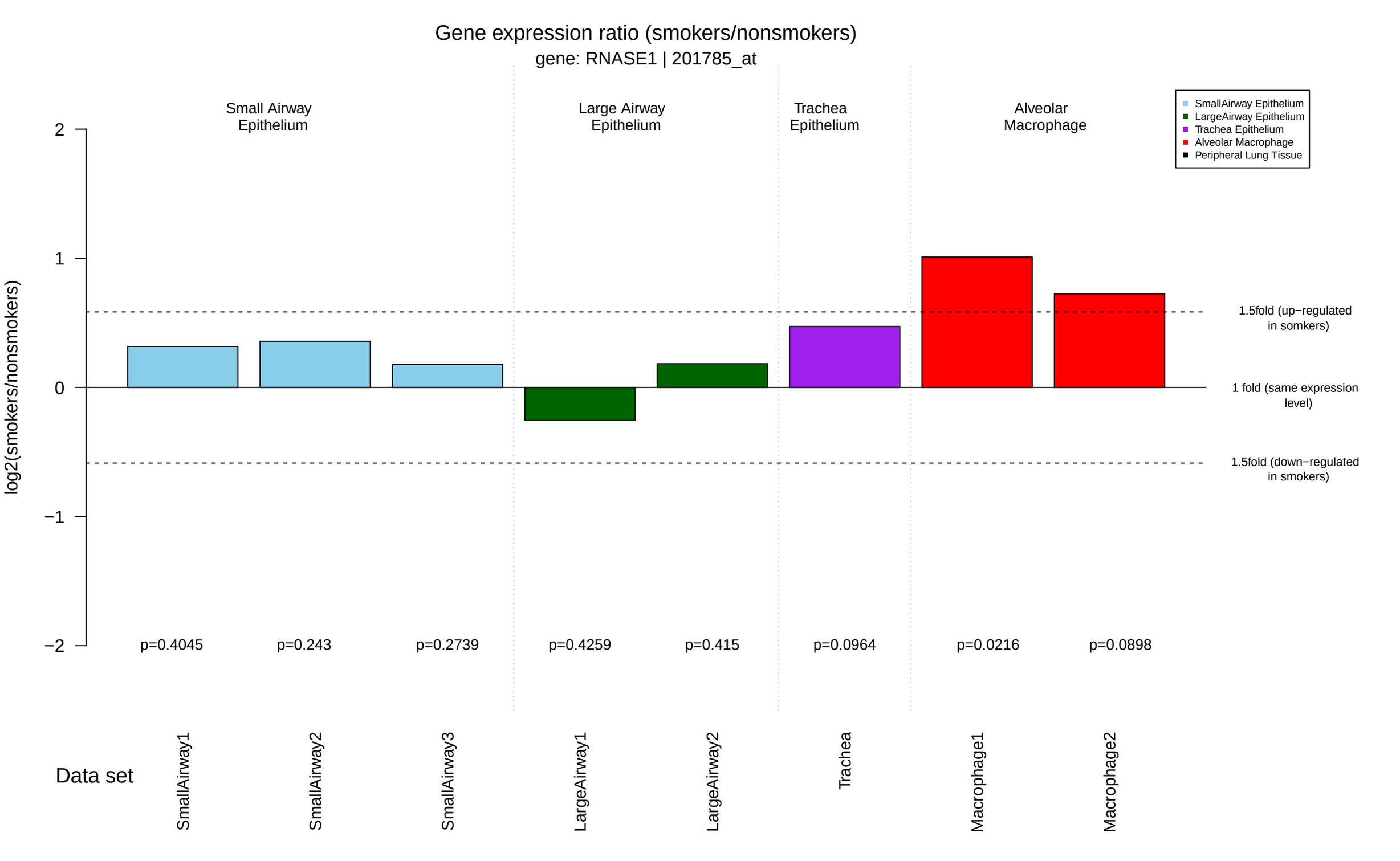
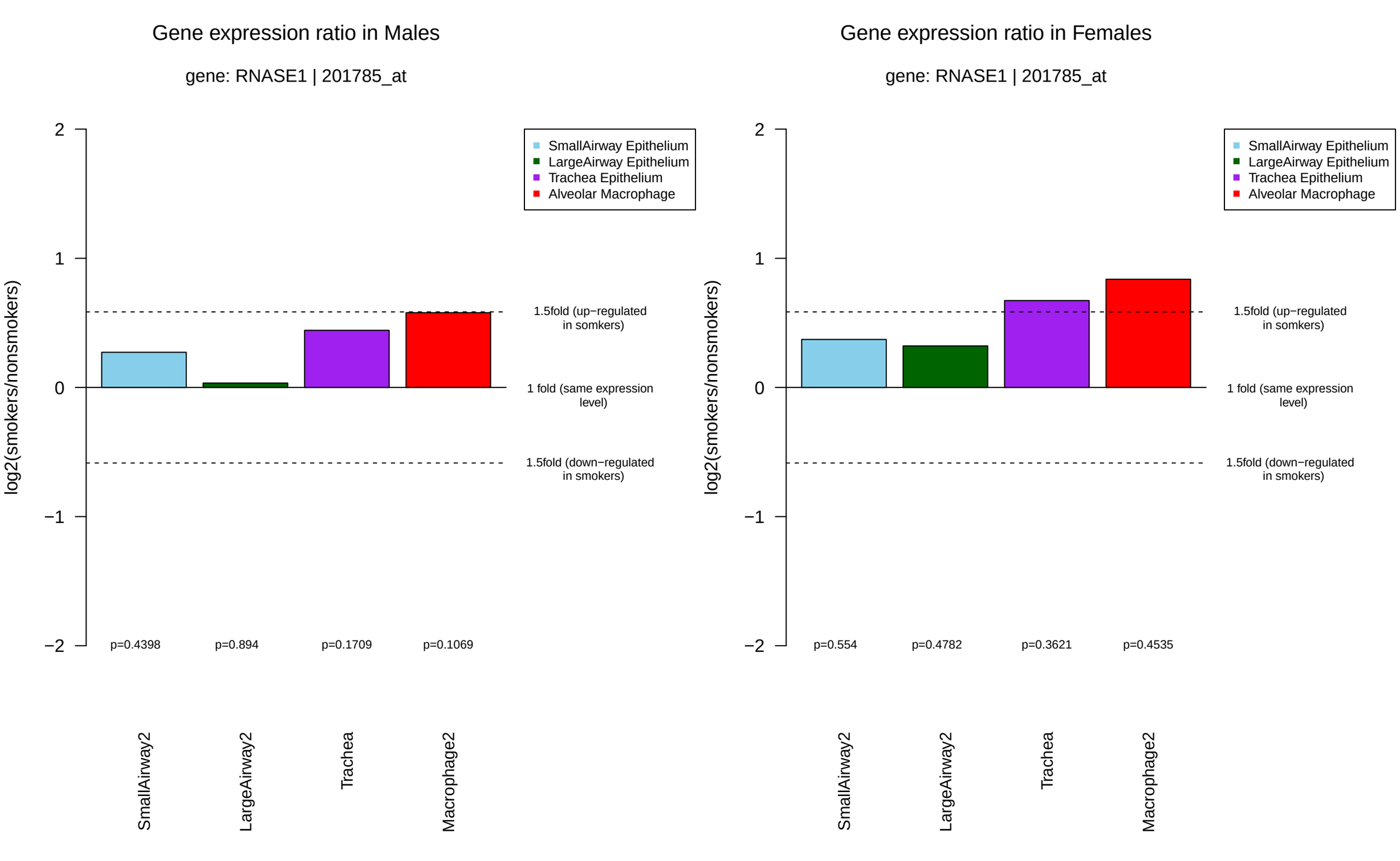
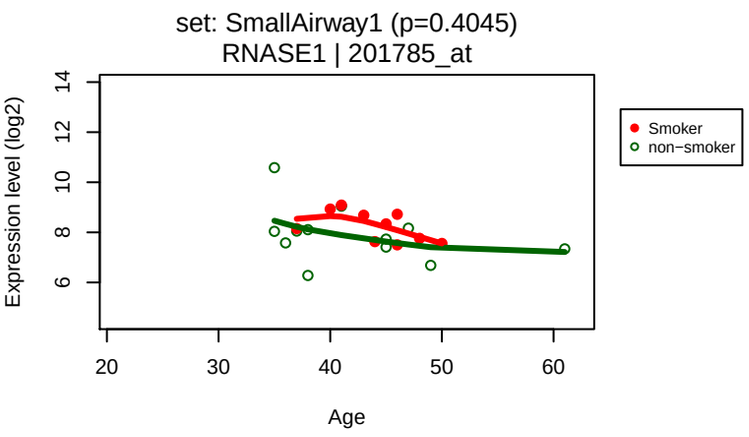
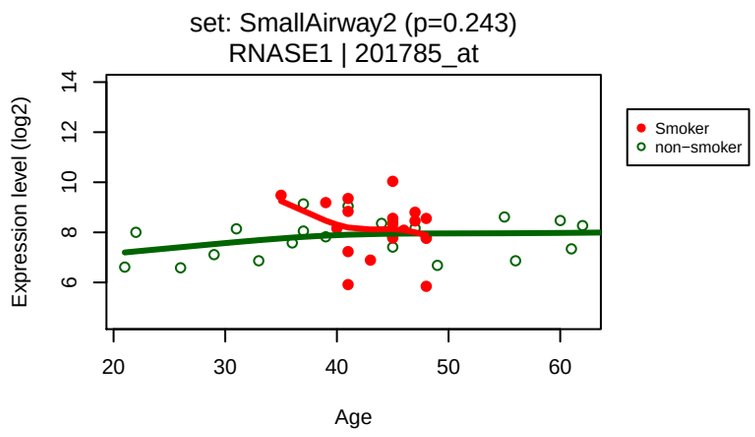
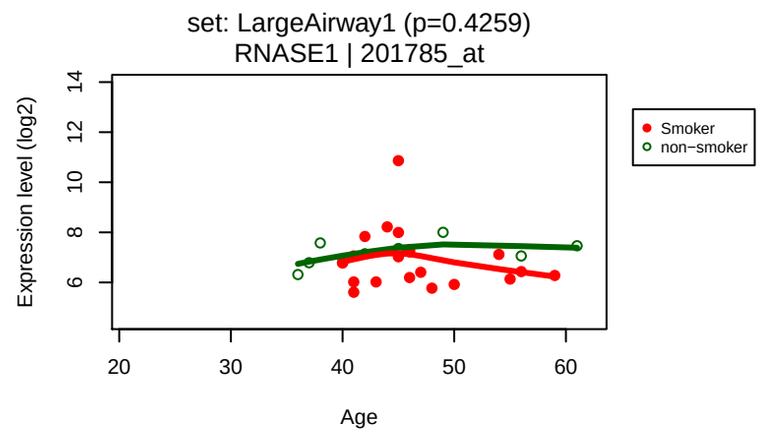

Gene expression of smokers and nonsmokers
gene: RNASE1 | 201785_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4045
p=0.243
p=0.2739
p=0.4259
p=0.415
p=0.0964
p=0.0216
p=0.0898
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

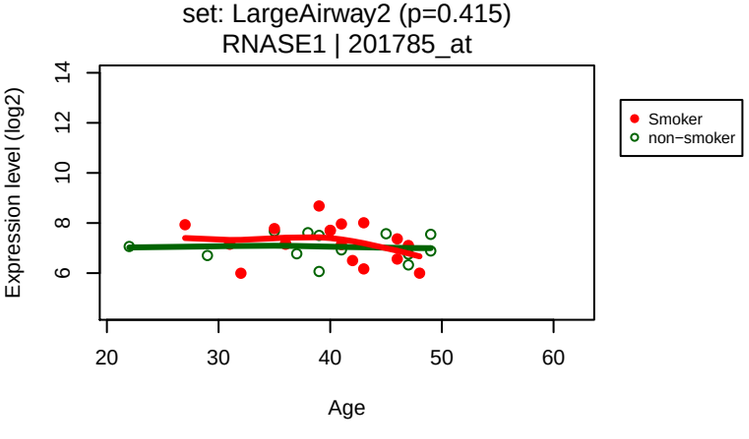

Gene expression ratio (smokers/nonsmokers)
gene: RNASE1 | 201785_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4045
p=0.243
p=0.2739
p=0.4259
p=0.415
p=0.0964
p=0.0216
p=0.0898
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

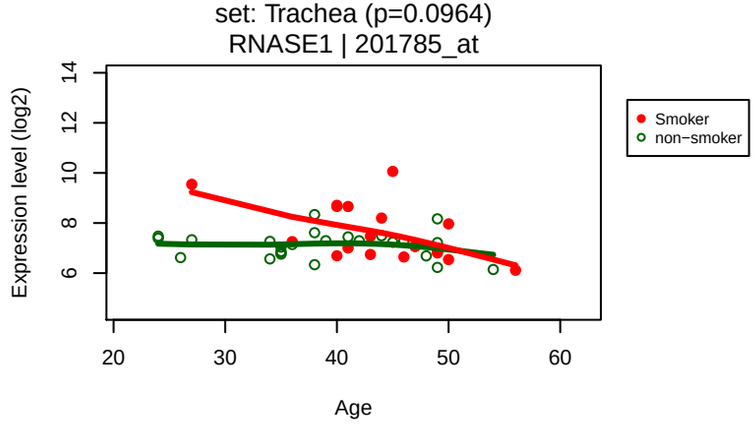

Gene expression ratio in Males
gene: RNASE1 | 201785_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4398
p=0.894
p=0.1709
p=0.1069
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RNASE1 | 201785_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.554
p=0.4782
p=0.3621
p=0.4535
SmallAirway2
LargeAirway2
Trachea
Macrophage2

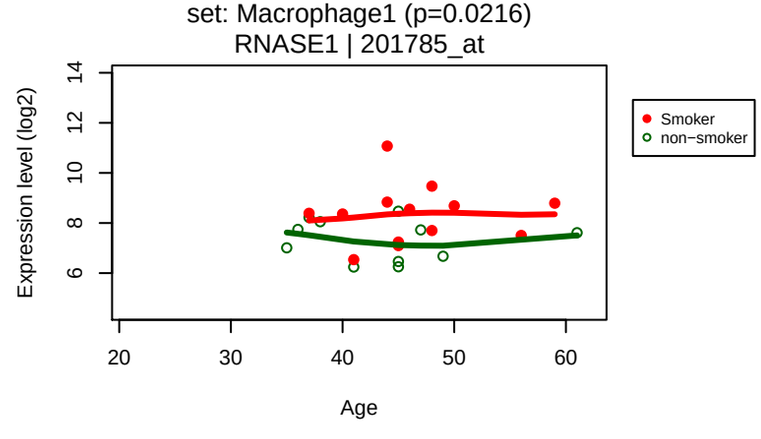

set: SmallAirway1 (p=0.4045)
RNASE1 | 201785_at
Expression level (log2)
Age
Smoker
non−smoker

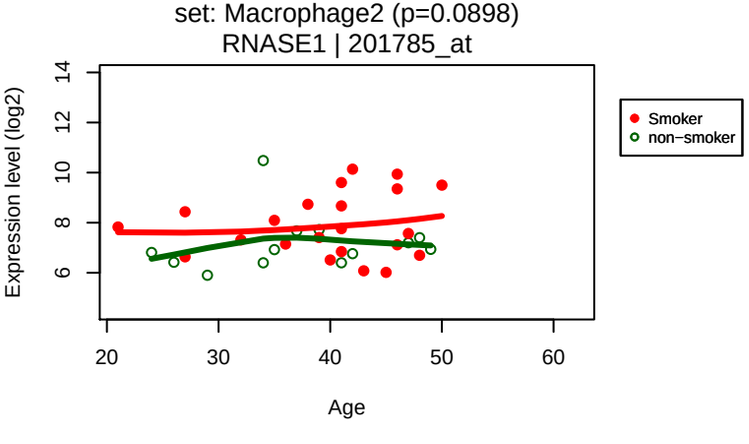

set: SmallAirway2 (p=0.243)
RNASE1 | 201785_at
Expression level (log2)
Age
Smoker
non−smoker

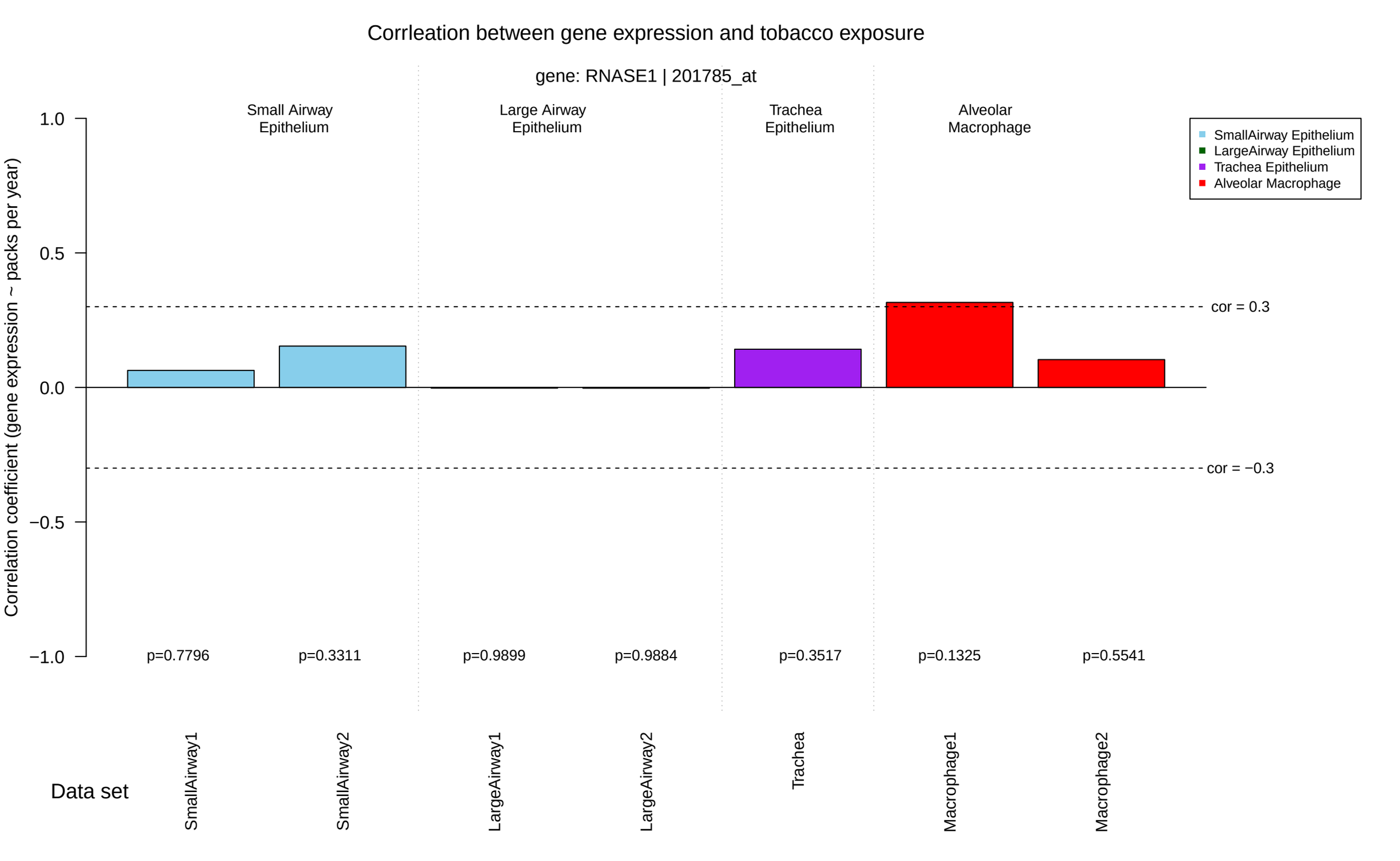

set: LargeAirway1 (p=0.4259)
RNASE1 | 201785_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.415)
RNASE1 | 201785_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0964)
RNASE1 | 201785_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0216)
RNASE1 | 201785_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0898)
RNASE1 | 201785_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RNASE1 | 201785_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7796
p=0.3311
p=0.9899
p=0.9884
p=0.3517
p=0.1325
p=0.5541
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2